<table>
<tr><th>User Story</th><th>L1 – Raspberry Pi</th><th>to</th><th>Week 47</th></tr>
<tr><td rowspan="3">Informal description</td><td colspan="3">As a technician</td></tr>
<tr><td colspan="3">... I would like to install and configure a Raspberry Pi</td></tr>
<tr><td colspan="3">... so that sensors can be read and their values displayed on the screen and sent to a web server.</td></tr>
<tr><td>Requirements according to the work order</td><td colspan="3">Must
• The Raspberry Pi is mounted on the screen and the screen is connected.
• The "Raspberry Pi OS" operating system is installed on the Raspberry Pi.
• The screen orientation (landscape) is set correctly.
• A static IP address (Ethernet) and a host name (IAM code) are configured.
• The I²C interface is enabled.
• The supplied files have been copied to the correct folders and the configuration file has been adjusted.
• The firmware starts automatically during boot.
• Docker is installed.
• The supplied website for displaying the current measurements is hosted by a web server in a Docker container.
• The website is displayed in kiosk mode.
• <u>Documentation:</u> The relevant installation and configuration steps with all necessary commands and explanations.
Should
• The Raspberry Pi is accessible via SSH.
• The current measurements are displayed in an appealing manner.
Could
• WLAN is set up and configured correctly.
• SSH access is not possible with a password.
• The measurement data history is displayed.</td></tr>
<tr><td colspan="4" align="center">Tasks according to the work order</td></tr>
</table>

- Install "Raspberry Pi OS (64-bit)" on the microSD card.
- Configure the operating system according to specifications.
- Activate the I^2C interface [1].
- Install Docker using the script (get.docker.com, [2]).
- Create the folders `/opt/station` and `/var/station`. Ensure that the permissions are correct.
- Copy the supplied files (firmware including configuration file, website, and scripts for testing the sensors) to the Raspberry Pi.
    - All files for the website are located in the `/var/www/html` folder.
    - All firmware files are located in the `/opt/station` folder.
- Adjust the configuration file according to the work order. (In particular, the serial number and I^2C addresses will be configured later in the Atelier User Stories.)
- Create a Docker container (php:8.4-apache) [3]:
```
$ docker run -d -name php -p 8080:80 -v /tmp/station:/tmp/station -v opt/station/www:/var/www/html php:8.4-apache
```
- Configure autostart [4].
  Use the file `~/.config/labwc/autostart` for this. This is executed as a shell script after startup.
  Note: Commands that run for a long time must be started in the background!
    - Create the folder `/tmp/station`.
    - Start the script (firmware) for reading and sending the sensors.
      Attention: This is a Python3 script!
    - Start the container (name: php) automatically.
      The supplied website is now accessible at http://localhost:8080.
    - Automatically start the browser in full-screen mode (kiosk) and display the current measured values.

Resources

- [1] https://www.raspberrypi.com/documentation/computers/configuration.html#raspi-config
- [2] https://docs.docker.com/engine/install/debian/
- [3] https://docs.docker.com/engine/storage/bind-mounts/
- [4] https://www.raspberrypi.com/tutorials/how-to-use-a-raspberry-pi-in-kiosk-mode/